DATA ANALYTICS PROJECT REPORT

House Price Analysis

Exploring property prices in relation to area, bedrooms, bathrooms and property type

Project Details	Information
Project	House Price Analysis
Dataset	house_prices.csv
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook
Report Type	Academic and Business Analysis Report

Prepared for academic and professional portfolio use

EXECUTIVE SUMMARY

This project examines the factors associated with house prices in a dataset of 300 records and 8 features. The planned analysis focuses on the relationship between property area and price, the role of bedrooms and bathrooms, differences by property type and the overall distribution of prices. The report provides a framework for interpreting the dataset and translating results into property-market insights. Since no calculated price statistics or correlation coefficients were supplied, numerical conclusions are intentionally left for verification from the notebook.

1. INTRODUCTION

House-price analysis is useful to buyers, sellers, property consultants and market researchers. Property prices may vary with size, layout, property type and other characteristics. Exploratory data analysis helps identify patterns, unusual observations and possible relationships before any predictive model is developed. This project uses descriptive and visual analysis to study the supplied property dataset.

2. PROJECT OBJECTIVES

- Describe the distribution of house prices.
- Examine the relationship between property area and price.
- Compare prices across bedroom and bathroom counts.
- Analyse price differences among property types.
- Identify relationships among numeric property variables.
- Develop cautious, data-supported observations for buyers, sellers and property analysts.

3. DATASET DESCRIPTION

The house-prices dataset is specified as containing 300 property records and 8 features. It is intended to include a target price field and property characteristics such as area, number of bedrooms, number of bathrooms and property type. Confirm exact field names, units and category definitions from the CSV before analysis.

Attribute	Description
Dataset file	house_prices.csv
Records	300
Features	8
Analysis environment	Jupyter Notebook with Python libraries

4. DATA PREPARATION AND VALIDATION

Load the CSV and inspect the structure, data types and summary statistics. Check for missing values, duplicate properties and inconsistent property-type labels. Ensure price and area are numeric and use consistent units, such as square feet or square metres. Check for impossible or suspicious values, including non-positive prices or areas. Inspect extreme prices and areas using box plots and quantiles; do not remove outliers automatically because premium properties may be genuine. Confirm whether bedroom and bathroom counts are stored as integers and whether property type categories are consistently labelled.

5. EXPLORATORY AND STATISTICAL ANALYSIS

Begin with measures of central tendency and spread for price and area: mean, median, standard deviation, minimum, maximum and quartiles. Use a histogram to understand price distribution and a box plot to identify potential outliers. Calculate Pearson correlation for approximately linear relationships between numeric variables and consider Spearman correlation when relationships are monotonic but not linear or when outliers are influential. For categorical property type, compare group counts and price distributions. Any regression model should be evaluated with a train/test split and suitable metrics such as MAE, RMSE and R-squared; correlation alone is not a predictive model.

6. VISUALIZATION AND INTERPRETATION

Visualization	Purpose and interpretation
Price Distribution	A histogram and box plot show the typical price range, skewness and potential outliers. Report mean and median together because property prices are often influenced by high-value properties.
Area vs Price	A scatter plot helps assess whether larger properties tend to have higher prices. Add a trend line only when it is appropriate and describe the strength of the relationship using a verified correlation or model.
Bedrooms vs Price	A box plot or grouped bar chart compares price distributions by bedroom count. Group sample sizes should be displayed to avoid overinterpreting categories with few properties.
Property Type Analysis	Compare median and mean prices across property types. Use consistent categories and identify whether differences may reflect area or location composition.
Correlation Heatmap	Display correlations among numeric variables such as price, area, bedrooms and bathrooms. Correlation measures association, not causation, and may be affected by outliers.

7. KEY FINDINGS

- The dataset contains 300 records and 8 features, according to the supplied specification.

- The planned analysis includes price distribution, area-price relationship, bedroom and bathroom comparisons, property-type analysis and a correlation heatmap.
- No exact mean, median, price range, correlation coefficient or property-type price ranking was supplied.

8. BUSINESS INSIGHTS

Area, bedroom count and bathroom count are useful property descriptors, but their relationships with price must be verified from the data. A positive area-price association, if observed, would suggest that larger properties tend to command higher prices in this sample; it would not prove that area alone determines price. Property-type comparisons can reveal market segments, but differences may also reflect unobserved factors such as location, age, condition and amenities.

9. BUSINESS RECOMMENDATIONS

- Use median price as well as mean price when summarizing a market that may contain premium outliers.
- Compare properties of similar area and type before drawing conclusions about bedroom or bathroom effects.
- Record and standardize area units before calculating price per unit area.
- Investigate unusual observations against the original records rather than deleting them automatically.
- Collect additional variables such as location, property age, condition and amenities to improve future analysis.
- Treat any predictive model as an estimate and evaluate it on held-out data before practical use.

10. LIMITATIONS AND FURTHER WORK

The supplied summary did not include exact prices, average area, median price, correlations or model performance. Consequently, this report does not state unsupported market averages or claim that any feature is the strongest price predictor. A dataset of 300 records may not represent an entire city or region. Results are descriptive and depend on the sample, data quality, property mix and time period.

11. CONCLUSION

House Price Analysis organizes property data into measurable comparisons and helps reveal how prices vary with property characteristics. The planned charts and statistical summaries can support buyers and analysts in understanding the sample market. Before using the findings for valuation or investment decisions, verify all calculated statistics and consider additional explanatory variables and a larger, more representative dataset.

APPENDIX: REPRODUCIBILITY CHECKLIST

- Retain the original dataset and the cleaned dataset separately.
- Record the exact cleaning steps and the number of rows removed or modified.

- Run all notebook cells from top to bottom before exporting the final notebook.
- Label every chart with a clear title, axis names and units.
- Add exact calculated values to the report only after verifying them from the notebook output.
- Keep the analysis code, report and README together in the project repository.